

Team Responsibilities

Comprehensive AI-Powered Dental Diagnostic
& Treatment System with Multi-Condition Detection

Team Members Overview

Student ID	Name	Module Assigned
IT24104023	Zakee M.N.M	User Profile Management
IT24103679	Kumarasiri A.Y.D	Doctor's Digital Prescription Module
IT24101772	Thathsala I.G.D	Medication Ordering & E-Payment Gateway
IT24100052	Ahamed M. S. A	Comprehensive Medical Resource & History Vault
IT24100022	Pavesh T.	Feedback and Rating
IT24103679	Ilham M. H. M	Appointment Scheduling
IT24103652	Jayasooriya G.D.R.M.	Emergency Contact Management

Zakee M.N.M

IT24104023 · User Profile Management

Responsible for the complete user authentication and profile management system. Zakee built the JWT-based security layer that protects all private API routes across the application. He also configured the MongoDB Atlas database connection and the Node.js HTTP server, forming the backbone that all other modules depend on. His work includes user registration, login, profile update, and account deactivation flows.

Backend Files	Frontend Files / Screens
- config/db.js - models/User.js - controllers/authController.js - middleware/authMiddleware.js - routes/authRoutes.js - server.js .env configuration	- screens/auth/LoginScreen.js - screens/auth/RegisterScreen.js - screens/auth/ProfileScreen.js - screens/auth/EditProfileScreen.js - services/api.js (auth token setup) - navigation/AppNavigator.js

Notable Features

- Password hashing using bcryptjs before saving to the database
- JWT token generation, expiry management, and verification on all protected routes
- Role-based access control differentiating patient, doctor, and admin roles
- Profile image upload linked to user account record
- Account deactivation and deletion with cascading record updates

Kumarasiri A.Y.D

IT24103679 · Doctor's Digital Prescription Module

Built the complete digital prescription module covering both backend API and frontend screens. This module allows dentists to create structured digital prescriptions with selected medications and dosages, modify them before final submission, and delete drafts. Patients can view their full prescription history through the patient portal. The prescription model serves as the link between the consultation and the medication ordering module.

Backend Files	Frontend Files / Screens
• <code>models/Prescription.js</code> • <code>controllers/prescriptionController.js</code> • <code>routes/prescriptionRoutes.js</code>	• <code>screens/prescriptions/PrescriptionCreateScreen.js</code> • <code>screens/prescriptions/PrescriptionListScreen.js</code> • <code>screens/prescriptions/PrescriptionDetailsScreen.js</code> • <code>screens/prescriptions/PrescriptionDraftScreen.js</code>

Notable Features

- Structured medication entries with name, dosage, frequency, and duration fields
- Draft state allows the prescribing doctor to edit before final submission
- Prescription lock on medication order creation — prevents editing after an order is placed
- Patient-facing prescription history with doctor name and consultation date
- Doctor-only delete permission enforced via role header check

Thathsala I.G.D

Built the medication ordering and mock e-payment module. Patients can create medication orders directly from approved prescriptions, submit card payment details which are validated for correct format only, and track their order status from pending through to confirmed. The admin verification dashboard allows administrators to approve or reject pending payment requests. No real payment gateway is integrated in this academic phase — card format validation simulates the full user experience.

Backend Files	Frontend Files / Screens
- models/MedicationOrder.js - models/Payment.js - controllers/orderController.js - controllers/paymentController.js - routes/orderRoutes.js - routes/paymentRoutes.js	- screens/orders/OrderScreen.js - screens/orders/OrderHistoryScreen.js - screens/payment/PaymentScreen.js - screens/payment/PaymentHistoryScreen.js - screens/admin/AdminOrderVerificationScreen.js

Notable Features

- Card number, expiry date (MM/YY), CVV, and cardholder name validated for correct format
- Card number masked to **** * before storing — CVV is never stored
- Order status lifecycle: Pending > Pending Admin Approval > Confirmed or Rejected
- Admin verification dashboard showing all pending orders with full prescription and payment details
- Unpaid and pending orders can be cancelled by the patient at any time
- Refund processing updates order, payment, and ticket status simultaneously

Ahamed M. S. A

IT24100052 · Comprehensive Medical Resource & History Vault

Built the central medical record management module — the Comprehensive Medical Resource and History Vault. This module stores all patient records including X-rays, intraoral photos, clinical reports, doctor schedules, and inventory entries. It serves as the data backbone of the system and bridges directly with the AI/ML diagnostic engine via the file_url field, allowing the Multi-Condition Dental Diagnostic Engine (MC-DDE) to store and retrieve diagnostic images through this vault. All deletions are soft-deletes to preserve a full audit trail.

Backend Files	Frontend Files / Screens
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- `models/MedicalRecord.js`
- `controllers/medicalRecordController.js`
- `routes/medicalRecordRoutes.js`
- `middleware/authMiddleware.js`
- `config/db.js`
- `server.js`

- `screens/records/RecordListScreen.js`
- `screens/records/AddRecordScreen.js`
- `screens/records/EditRecordScreen.js`
- `screens/records/RecordDetailScreen.js`
- `screens/records/PatientHistoryScreen.js`

Notable Features

- Five record types supported: xray, photo, report, schedule, inventory — each with distinct UI badge colours
- Soft-delete only (`is_deleted: true`) — records are never physically removed; full audit trail is preserved
- Admin-only delete enforced via `x-user-role: admin` request header; returns 403 Forbidden for unauthorized attempts
- `file_url` field bridges the vault with the MC-DDE AI engine — xray and photo records are used for inference
- Patient history retrieved sorted by `record_date` descending with pagination (10 records per page)
- All required fields validated server-side with descriptive 400 error messages listing missing fields

Pavesh T.

IT24100022 · Feedback and Rating

Built the feedback and rating module that allows patients to submit star ratings and written comments after completed consultations. Feedback can be edited within a policy-defined time window and deleted by the patient. Doctors can view their aggregate rating summary and individual feedback entries through the doctor portal. The module integrates with the appointment system to verify that feedback is linked to a completed consultation.

Backend Files	Frontend Files / Screens
• <code>models/Feedback.js</code> • <code>controllers/feedbackController.js</code> • <code>routes/feedbackRoutes.js</code>	• <code>screens/feedback/FeedbackScreen.js</code> • <code>screens/feedback/FeedbackHistoryScreen.js</code> • <code>screens/feedback/RatingSummaryScreen.js</code> • <code>screens/feedback/EditFeedbackScreen.js</code>

Notable Features

- Star rating from 1 to 5 with written comment — both stored and displayed per consultation
- Edit window enforced: feedback can only be modified within the allowed time period after submission
- Doctor rating summary screen aggregates all ratings with average score and recent comment history
- Feedback linked to `appointment_id` to ensure only completed consultations can be rated
- Patient feedback history screen showing all submitted ratings with statuses

Ilham M. H. M

IT24103652 · Appointment Scheduling

Built the appointment scheduling module covering both the backend booking API and the frontend calendar screens. The system allows patients to book consultations with available doctors by selecting a date and time slot from an interactive calendar. It prevents double bookings by checking for time-slot conflicts before confirming any appointment. Appointments align with Medical Record dates, enabling seamless patient history tracking across the system.

Backend Files	Frontend Files / Screens
- models/Appointment.js - controllers/appointmentController.js - routes/appointmentRoutes.js	- screens/appointments/AppointmentScreen.js - screens/appointments/BookingCalendarScreen.js - screens/appointments/AppointmentHistoryScreen.js - screens/appointments/AppointmentDetailScreen.js

Notable Features

- Interactive availability calendar with colour-coded slots: Green (available), Red (confirmed), Orange (pending)
- Backend time-slot overlap detection prevents double bookings before saving
- Rescheduling is permitted only within the system-defined advance notice window
- Appointment status lifecycle: Pending > Confirmed > Completed or Cancelled
- Patient appointment history shows both upcoming and past consultations with real-time status

Jayasooriya G.D.R.M.

IT24103652 · Emergency Contact Management

Built the emergency contact management module, including the AI/ML contribution for Predictive Analytics and Generative Simulation. The emergency contact module allows patients to register, update, and remove emergency contacts with name, relationship, phone number, and a primary contact flag. The AI/ML work covers simulating disease progression over six months, visualizing expected treatment outcomes, and generating automated procedure code recommendations from diagnostic results.

Backend Files	Frontend Files / Screens
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- `models/EmergencyContact.js`
- `controllers/emergencyContactController.js`
- `routes/emergencyContactRoutes.js`
- `ai/predictiveAnalytics.py`
- `ai/treatmentSimulation.py`

- `screens/emergency/EmergencyContactScreen.js`
- `screens/emergency/AddContactScreen.js`
- `screens/emergency/EditContactScreen.js`
- `screens/ai/ProgressionSimulationScreen.js`
- `screens/ai/TreatmentOutcomeScreen.js`

Notable Features

- Primary contact flag: one contact per patient can be marked as the priority emergency contact
- Full CRUD: add, view, update, and remove contacts — all linked by `patient_id`
- Disease progression simulation: shows how untreated caries or calculus worsens over six months
- Treatment outcome visualization: renders simulated post-treatment result for patient motivation
- Automated treatment planning: maps AI detection results to specific dental procedure codes and cost estimates

Module-to-File Quick Reference

Module	Student ID	Backend Files	Frontend Screens
Auth & User Profile	IT24104023	<code>models/User.js</code> <code>controllers/authController.js</code> <code>routes/authRoutes.js</code> <code>middleware/authMiddleware.js</code> <code>config/db.js</code> <code>server.js</code>	<code>LoginScreen.js</code> <code>RegisterScreen.js</code> <code>ProfileScreen.js</code> <code>EditProfileScreen.js</code>
Prescription Module	IT24103679	<code>models/Prescription.js</code> <code>controllers/prescriptionController.js</code> <code>routes/prescriptionRoutes.js</code>	<code>PrescriptionCreateScreen.js</code> <code>PrescriptionListScreen.js</code> <code>PrescriptionDetailScreen.js</code>
Medication & Payment	IT24101772	<code>models/MedicationOrder.js</code> <code>models/Payment.js</code> <code>controllers/orderController.js</code> <code>routes/orderRoutes.js</code>	<code>OrderScreen.js</code> <code>OrderHistoryScreen.js</code> <code>PaymentScreen.js</code>
Medical Record Vault	IT24100052	<code>models/MedicalRecord.js</code> <code>controllers/medicalRecordController.js</code> <code>routes/medicalRecordRoutes.js</code> <code>middleware/authMiddleware.js</code>	<code>RecordListScreen.js</code> <code>AddRecordScreen.js</code> <code>EditRecordScreen.js</code> <code>RecordDetailScreen.js</code>
Feedback & Rating	IT24100022	<code>models/Feedback.js</code> <code>controllers/feedbackController.js</code> <code>routes/feedbackRoutes.js</code>	<code>FeedbackScreen.js</code> <code>FeedbackHistoryScreen.js</code> <code>RatingSummaryScreen.js</code>
Appointment Scheduling	IT24103679	<code>models/Appointment.js</code> <code>controllers/appointmentController.js</code> <code>routes/appointmentRoutes.js</code>	<code>AppointmentScreen.js</code> <code>BookingCalendarScreen.js</code> <code>AppointmentHistoryScreen.js</code>

Module	Student ID	Backend Files	Frontend Screens
Emergency Contacts	IT24103652	models/EmergencyContact.js controllers/emergencyContactController.js routes/emergencyContactRoutes.js	EmergencyContactScreen.js AddContactScreen.js EditContactScreen.js